

Optimization

UChicago Political Science Math Camp

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Why Optimize?

Many (most) results in quantitative social sciences involve the “best” alternative given a set of possibilities and a payoff metric:

- ▶ **Formal Theory:** Almost exclusively interested in highest-payoff alternatives
- ▶ **Regression:** OLS generated by minimizing errors from linear formulas
- ▶ **Approximations:** Always interested in closest approximation given certain constraints

Optimization usually refers to extrema.

An Opening Question

What characteristics does a maximal (minimal) value have?

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What must be true of the relation between that value and every other element?

Mean Value Theorem

The **mean value theorem** states the following:

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists some $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

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At some point over a domain, a function's derivative must equal its overall slope over that domain.

An Important Corollary

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Suppose $a < b$ and $f(a) < f(b)$. Then, there must be some point $c \in [a, b]$ such that $f'(c) > 0$.

How can we apply this to our definition of extrema?

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How can we apply this to our definition of extrema?

Suppose $f'(c) \neq 0$ **for any range of** $[a, b]$. Then, $f(a) \neq f(b)$, so either $f(a) > f(b)$ or $f(a) < f(b)$. *Then, $f(c)$ cannot be an (interior) extrema.*

Critical Points

A critical point occurs at x when $f'(x) = 0$. We derive as follows:

$$f(x)$$
$$\frac{d}{dx}f(x)$$
$$x^* \implies \frac{d}{dx}f(x) = 0$$

This is called the **first order condition**, and is one of two places where an extrema can be found.

This Doesn't Get Us All The Way There

Let us solve the following:

$$\arg \min_x f(x) = x^2 : x \in [0, 2]$$

$$\arg \max_x f(x) = x^2 : x \in [0, 2]$$

End Points

Extrema can also be found at the extrema of a function's domain.

This means there are two steps to evaluating the extrema of some function f :

1. Identify critical points and evaluate f at each
2. Identify endpoints and evaluate f at each

Concavity, Quickly

*A function f is said to be **concave** over $[a, b]$ if every point $f(c) : c \in [a, b]$ lies above the linear combination of $f(a), f(b)$, and is **convex** if every point $f(c)$ lies below that line.*

How might we apply this definition to the second derivative?

Categorizing Critical Points

The nature of a critical point is determined by the function's **second derivative** at that point:

▶ $\frac{d^2f(x)}{dx^2} > 0 \implies$ **minima**

▶ $\frac{d^2f(x)}{dx^2} < 0 \implies$ **maxima**

▶ $\frac{d^2f(x)}{dx^2} = 0 \implies$ **saddle point**

Categorizing Critical Points

Extrema can take one of two forms, and often must be categorized:

1. **Global** extrema, such that no value in the domain exceeds it
2. **Local** extrema, such that some value in the domain does exceed it

Sometimes, your best bet is to just “plug-and-chug.” [When might this not be the case?](#)

Denoting Optimization: min/max

We use two operations for optimization: **min/max** and **argmin/argmax**

To refer to the minimal value of $f(x)$:

$$\min f(x)$$

To solve $\min f(x)$ over the domain $x \in [a, b]$:

1. Use the first order condition to find x^* and the second derivative to categorize as **min** or **max**; discard maxima (and saddle)

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2. Evaluate $f(x^*)$ at minima
3. Evaluate $f(x)$ at endpoints $x = \{a, b\}$
4. The minimal value of f over critical points and end points solves $\min f(x) : x \in [a, b]$

Denoting Optimization: $\operatorname{argmin}/\operatorname{argmax}$

To refer to the value of x which returns the minimal value of $f(x)$:

$$\operatorname{argmin}_x f(x)$$

Note that this usually requires a specific “w.r.t.” variable, but uses similar steps to solve $\operatorname{argmin}_x f(x) : x \in [a, b]$:

1. Use the first order condition to find x^* and the second derivative to categorize as **min** or **max**; discard maxima (and saddle)
2. Evaluate $f(x^*)$ at minima
3. Evaluate $f(x)$ at endpoints $x = \{a, b\}$
4. The minimal value of f over critical points and end points solves $\min f(x) : x \in [a, b]$
5. The value(s) of x which produce $\min f(x) : x \in [a, b]$ then satisfy $\operatorname{argmin}_x f(x) : x \in [a, b]$

Some Practice Problems

Solve both $\arg \max_x f(x)$ and $\max f(x)$ for the following functions and domains:

$$f(x) = x^2 + x \quad : \quad x \in [-2, 2]$$

$$f(x) = \frac{1}{1+x^2} \quad : \quad x \in \mathbb{R}$$

$$f(x) = \frac{1}{x^2} \quad : \quad x \in [a \in \mathbb{R}, b \in \mathbb{R}]$$

An Interesting Qualifier

A function can have multiple equivalent extrema (and therefore, multiple arguments that produce that value)

For that reason, though $\min f(x)$ and $\max f(x)$ return singular values, $\arg \min_x f(x)$ and $\arg \max_x f(x)$ return *sets*:

$$x^* \in \arg \max_x f(x)$$

Examples

Solve $\min f(x)$ and $\arg \min_x f(x)$ for the following:

$$f(x) = x^2 - 2x \quad : \quad x \in [0, 2]$$

$$f(x) = \frac{1}{1+x^2} \quad : \quad x \in \mathbb{R}$$

Monotonic Optimization

Sometimes, we can either solve or greatly simplify an optimization question by identifying **monotonic relationships**

In such cases, the following condition is satisfied:

$$\frac{d}{dx} f(x) \neq 0 \quad \forall x \in \mathbb{R}$$

Then, x is either *always maximized* or *always minimized*, and the optimization solution lies at an endpoint.

Monotonicity Example

Suppose we are considering the impact of legislature size on cleavage inclusion. Define:

- ▶ $L \in \mathbb{N}$: size of legislature
- ▶ $i \in \mathbb{N}$: the number of existing identity groups NOT represented in the legislature

What might we expect (or know) to be true about $\frac{\partial i}{\partial L}$? What bounds might exist, and what might this suggest about legislative inclusion?

The Difficult Part

is almost always figuring out *what* optimization to run, rather than *how* to run it

There are a few common tricks:

1. Posit the question w.r.t some metric
2. Instead of “What do I want to achieve?”, frame as “What do I want to optimize?”
3. Rearrange to identify monotonicity when possible

An Additional Consideration

So far, we have always worked with a clear variable of interest. But, we are often interested in multivariable functions...

Luckily, the pathway forwards is pretty simple.

Partial Derivatives: Step 1



Partial Derivatives: Step 2



Partial Derivatives

What happens to the gauge when you move one (1) lever?

This is contingent on two primary things:

1. What direction you move that lever
2. Where all the other levers are set

These conditions are re-evaluated at every point!

How Do We Approach This?

Multivariable calculus lends a very straightforward route for these solutions, both in differentiation and in integration:

Treat non-targeted variables as constants

Working Through This

Consider a very simple example:

$$f(x, y) = 2xy - x^2y^2$$

Then, we can solve:

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$$f(x, y) = 2xy - x^2y^2$$

Then, we can solve:

$$\begin{aligned}\frac{\partial f(x, y)}{\partial x} &= 2y - 2xy^2 \\ \frac{\partial f(x, y)}{\partial y} &= 2x - 2x^2y\end{aligned}$$

A Conceptual Question

What must be true in a multi-dimensional local maxima (or minima)?

Multivariate Optimization

Total optimization in multivariable environments follows a simple – even if technically exhausting – algorithm:

1. Find all critical points:

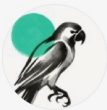
- 1.1 Solve $\frac{\partial f}{\partial x_i}$ for all x_i and identify FOCs

- 1.2 Solve FOCs as a system of equation to determine if any interior critical points exist

2. Find endpoint maxima / minima:

- 2.1 Optimize edge values

3. Compare values of all maxima / minima to determine extrema



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X.com

Local minima are extremely rare in high dimensional spaces, so if you ever feel stuck in a rut it's probably just because you aren't considering a wide enough set of orthogonal options

7:35 AM · 7/21/26 · **1.3M** Views



109



1K



11K



2.5K



A Very Common Optimization

From above: OLS = “sum of least squares” ...

That's just optimization!

Given $a < b < c$ and $f(a) < f(b) < f(c)$, **AND** with fixed $\hat{f}(0) = 0$ ¹, how might we optimize our regression estimator (linear slope)?

¹you're welcome!

OLS Optimization

OLS solves the following:

$$\min \sum \left(\hat{f}(a) - f(a) \right)^2 \quad \Rightarrow \quad \arg \min_m \sum \left(f(m, a) - f(a) \right)^2$$

We are holding $\hat{f}(0) = 0$, so $\hat{f}(x) = mx$. Then, we want to solve:

$$\arg \min_m \sum_i \left(m \cdot x_i - f(x) \right)^2$$

Expanded:

$$\arg \min_m \left[\left(ma - f(a) \right)^2 + \left(mb - f(b) \right)^2 + \left(mc - f(c) \right)^2 \right]$$

OLS Optimization

$$\begin{aligned} & \frac{d}{dm} \left[\left(ma - f(a) \right)^2 + \left(mb - f(b) \right)^2 + \left(mc - f(c) \right)^2 \right] \\ &= 2a \left(ma - f(a) \right) + 2b \left(mb - f(b) \right) + 2c \left(mc - f(c) \right) \end{aligned}$$

$$\frac{d}{dm} = 0 \implies 2a^2m + 2b^2m + 2c^2m = 2af(a) + 2bf(b) + 2cf(c)$$

$$2m(a^2 + b^2 + c^2) = 2 \left(af(a) + bf(b) + cf(c) \right)$$

$$m^* = \frac{af(a) + bf(b) + cf(c)}{a^2 + b^2 + c^2}$$

With One Little Difference

Now we know multivariable optimization. That's good!

Given $\left\{ \left(a, f(a) \right), \left(b, f(b) \right), \left(c, f(c) \right) \right\}$ and **without** any assumption on $\hat{f}(0)$, let's derive an optimal OLS estimator (both coefficient and slope).

This only needs the sum rule and power rule!

The Math

Define $\hat{f}(x) \equiv mx + z$ for $m, z \in \mathbb{R}$. Then, our optimization problem becomes:

$$\arg \min_{m,z} \left[\left(ma + z - f(a) \right)^2 + \left(mb + z - f(b) \right)^2 + \left(mc + z - f(c) \right)^2 \right]$$

$$\begin{aligned} \frac{\partial}{\partial z} &= 2ma + 2mb + 2mc - 2f(a) - 2f(b) - 2f(c) + 6z \\ &= 2m(a + b + c) - 2(f(a) + f(b) + f(c)) + 6z \end{aligned}$$

$$FOC : \frac{\partial}{\partial z} = 0 \implies z^* = \frac{-m(a + b + c) + f(a) + f(b) + f(c)}{3}$$

The Math

$$\frac{\partial}{\partial m} = 2m(a^2 + b^2 + c^2) + 2z(a + b + c) - 2\left(af(a) + bf(b) + cf(c)\right)$$

$$FOC : \frac{\partial}{\partial m} = 0 \implies m^* = \frac{af(a) + bf(b) + cf(c) - z(a + b + c)}{a^2 + b^2 + c^2}$$

$$\left(m^*, z^*\right) = \left(\frac{af(a) + bf(b) + cf(c) - z(a + b + c)}{a^2 + b^2 + c^2}, \frac{-m(a + b + c) + f(a) + f(b) + f(c)}{3}\right)$$

The Math

We want to express this only in terms of a, b, c , so we substitute:

$$m^* = \frac{af(a) + bf(b) + cf(c) - \frac{-m^*(a+b+c)+f(a)+f(b)+f(c)}{3}(a+b+c)}{a^2 + b^2 + c^2}$$
$$\Rightarrow m^* = \frac{3(af(a) + bf(b) + cf(c)) - (a+b+c)(f(a) + f(b) + f(c))}{3(a^2 + b^2 + c^2) - (a+b+c)^2}$$

$$z^* = \frac{-m^*(a+b+c) + f(a) + f(b) + f(c)}{3}$$
$$\Rightarrow z^* = \frac{-\left(\frac{3(af(a)+bf(b)+cf(c)) - (a+b+c)(f(a)+f(b)+f(c))}{3(a^2+b^2+c^2) - (a+b+c)^2}\right)(a+b+c) + f(a) + f(b) + f(c)}{3}$$

We could simplify further, but have no need

Run A Test

Set the following three data points: $(1, 2)$, $(2, 3)$, $(3, 5)$. Then, evaluate:

$$(m^*, z^*) =$$